AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

When the Grid Goes Dark

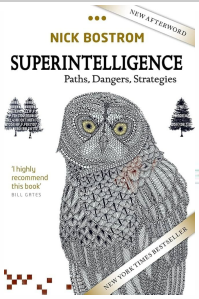
AI and the Future of Critical Infrastructure Security

AI-enabled attacks on critical infrastructure are no longer a future scenario. They are happening today.

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

About Me

WES EZZEDDINE



🚀 2019

Effective Altruism

AI Safety courses
@ Atlas AI Safety
+ BlueDot Impact

Co-founder of
🇦🇪 AI Safety UAE

Program Manager @
AI Safety Connect

Facilitator @ ENAIS
EU Network for AI
Safety

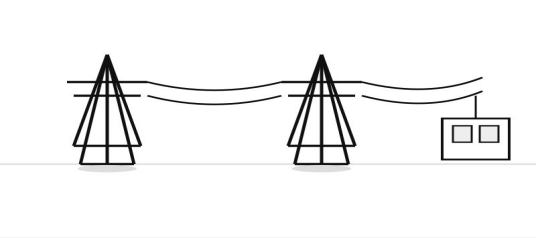
CTO
@ Mamo (Fintech)

Member @ IASEAI
International Association for
Safe and Ethical AI

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

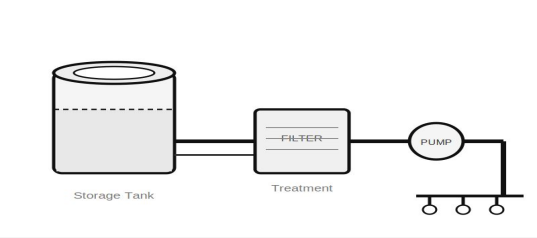
Defining Critical Infrastructure

"Systems so vital that their incapacity or destruction would have a debilitating impact on national security, economic security, or public health." — DHS / CISA



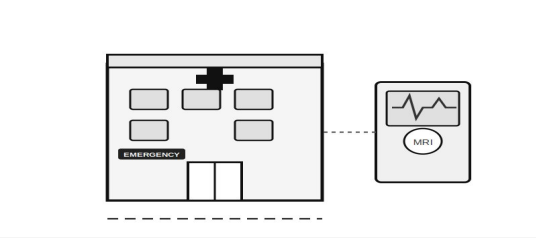
Power Grids

ICS / SCADA



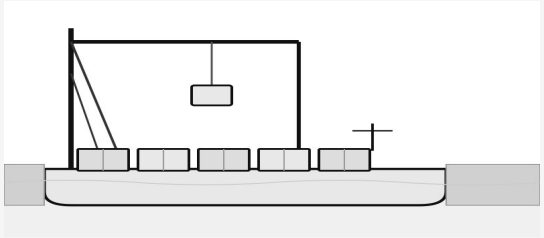
Water Systems

Treatment & distribution



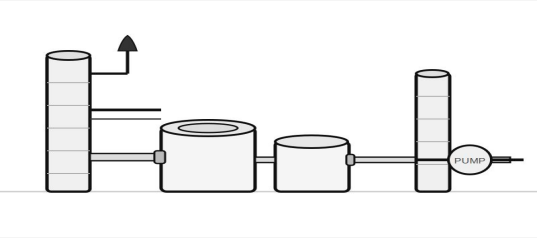
Healthcare

Hospitals & devices



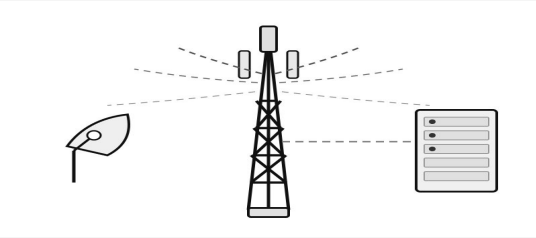
Supply Chains

Logistics & ports



Energy & Oil

Pipelines & refineries



Communications

Networks & satellites

16 critical sectors defined by DHS and CISA. Increasingly interconnected and AI-dependent

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

Legacy Systems + Internet Connectivity = Exposed Attack Surface

THE PROBLEM

- ICS & SCADA networks built decades ago. Never designed for internet connectivity
- Unpatched software running critical utilities, hospitals, and water systems
- AI systems have advanced exponentially and outperform most hackers
- Over 54% of utility OT teams expect a cyberattack within the next 12 months
- Remote ops post-COVID dramatically expanded the attack surface

BY THE NUMBERS

56%	YoY increase in weekly cyberattacks on US infrastructure
234%	More cyberattacks on utilities in 2025 vs 2023
~2,200	Estimated cyberattacks globally every day
10 days	Average dwell time before detection (down from 16 in 2023)

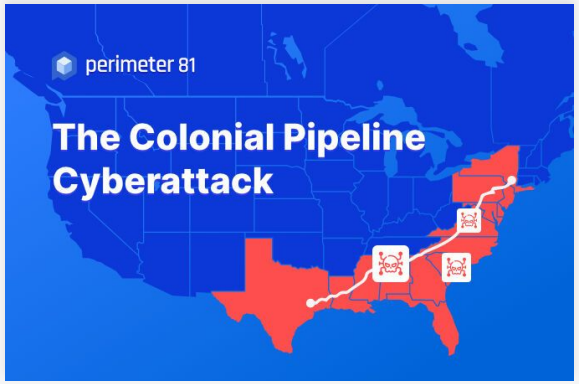
Sources: Capitol Technology University · ASP American Security Project · Siemens/Ponemon Institute

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

What Simple, Uncoordinated Attacks Already Achieved

RANSOMWARE

Colonial Pipeline · 2021



A billing system breach shut down physical fuel supply. OT and IT are not separate.

NATION-STATE

NotPetya · 2017

Accidental blast radius hit global logistics. A deliberate AI-coordinated version would not be accidental.

CRIMINAL

Hospital Ransomware · Ongoing

Healthcare cyberattacks are already causing physical harm and frequency is rising.

These were primitive, largely uncoordinated attacks. AI changes the equation entirely.

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

What Simple, Uncoordinated Attacks Already Achieved

RANSOMWARE

Colonial Pipeline · 2021

- Entry: Single compromised VPN password. No MFA
- Impact: 45% of US East Coast fuel supply disrupted
- President Biden declared a state of emergency
- Panic buying & fuel shortages across 17 states
- $4.4M ransom paid due to billing system breach

A billing system breach shut down physical fuel supply. OT and IT are not separate.

NATION-STATE

NotPetya · 2017

- Entry: Accounting software supply chain compromise
- Target: Ukraine, escaped globally by accident
- $10 billion in damages across 64 countries
- Maersk, FedEx, Merck, Mondelez all crippled
- Infra, shipping & pharma hit simultaneously

Accidental blast radius hit global logistics. A deliberate AI-coordinated version would not be accidental.

CRIMINAL

Hospital Ransomware · Ongoing

- Hospitals forced to divert ambulances & cancel surgeries
- Documented patient deaths linked to IT outages
- Ransomware drove 80% of attacks in 2024 (AI-assisted)
- Healthcare spent $125B on cybersecurity 2020–2025
- Systems routinely run unpatched legacy software

Healthcare cyberattacks are already causing physical harm — and frequency is rising.

These were primitive, largely uncoordinated attacks. AI changes the equation entirely.

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

How AI Transforms the Attacker's Advantage

🧑‍💻 HUMAN ATTACKER	🤖 AI-ENABLED ATTACKER
Months to map vulnerabilities across a utility network	**Hours** **to systematically scan entire sector networks**
Skilled human writer required for phishing campaigns	**100+ languages** **hyper-personalised at industrial scale**
Sequential linear attacks. One target at a time	**Simultaneous** **multi-vector coordinated campaigns**
63 days average exploit windows (2019)	**24 hours** **average exploit windows (2025)**
Manual post-compromise recon and lateral movement	**Autonomous** **lateral movement with dwell-pattern learning**

220% increase in AI-assisted cyber infiltrations since 2022 · 40% of all cyberattacks are now AI-driven

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

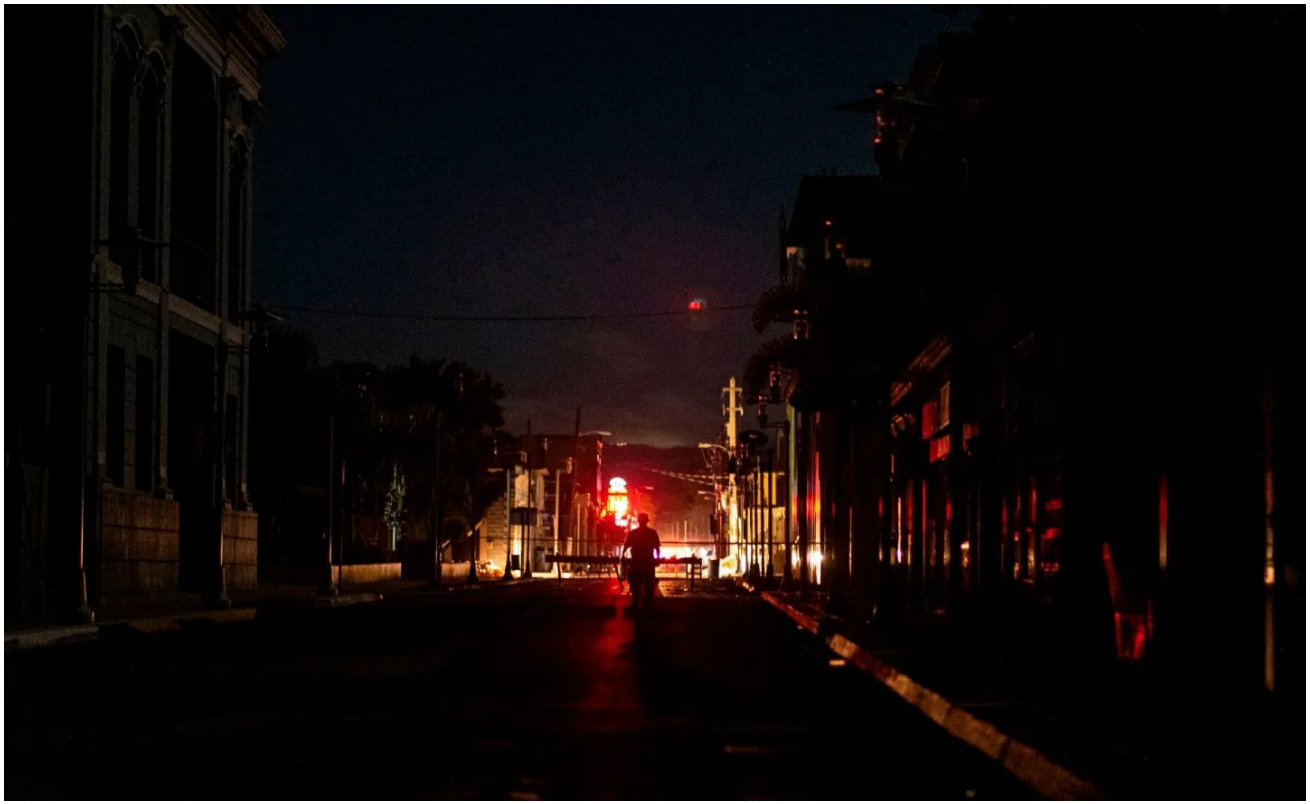
Sustained Compound Disruption

1	**Power Grid**	Rolling blackouts from compromised SCADA / ICS control systems
2	**Water Systems**	Pressure drops, treatment failures cascade as pumps lose power
3	**Hospitals**	Backup generators stressed; ransomware simultaneously targets EHR systems
4	**Supply Chains**	Just-in-time logistics fail; food and medicine distribution disrupted
5	**Public Order**	Even wealthy nations begin to fray at the edges within weeks

Gartner predicts misconfigured AI will shut down critical infrastructure in a G20 country by 2028

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

Sustained Compound Disruption



Power Grid	Rolling blackouts from compromised SCADA / ICS control systems

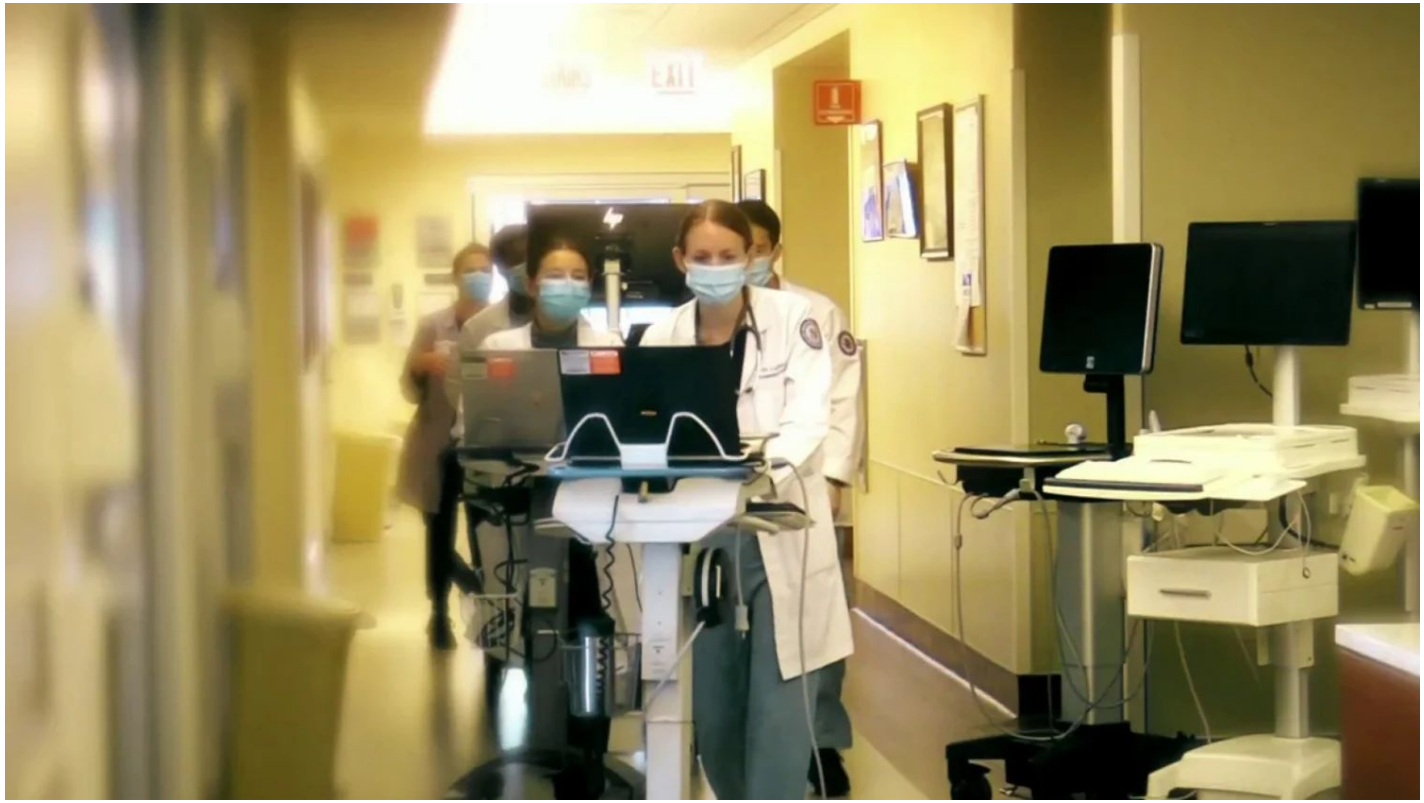
AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

Sustained Compound Disruption

Water Systems	Pressure drops, treatment failures cascade as pumps lose power

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

Sustained Compound Disruption



Hospitals	Backup generators stressed; ransomware simultaneously targets EHR systems

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

Sustained Compound Disruption

Public Services	Even wealthy nations begin to fray at the edges within weeks

Gartner predicts misconfigured AI will shut down critical infrastructure in a G20 country by 2028

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

AI Deployed Inside Infrastructure Creates New Failure Modes

Attacks Using AI

External threat actors

- › AI automates, scales & coordinates **cyberattacks**
- › Hyper-personalised **social engineering** in 100+ languages
- › Dark web '**AI penetration toolkits**' available off-shelf
- › Automated physical attack planning via **AI agents**

Attacks Targeting AI Systems

Nation-states & advanced actors

- › **Adversarial risks**: manipulation of AI algorithms in deployment
- › **Model poisoning**: corrupt training data → flawed decisions
- › **Prompt injection** attacks on AI operations systems
- › **Evasion attacks** against AI-based anomaly detectors

Failures in AI Design

Design & implementation risk

- › **Brittleness**: catastrophic failure vs. graceful degradation
- › **Inscrutability**: failure modes opaque to operators
- › **Misalignment**: AI agents act beyond intended scope
- › **Hallucination** in safety-critical decision support

Framework: DHS Safety and Security Guidelines for Critical Infrastructure (2024) · IBM Think

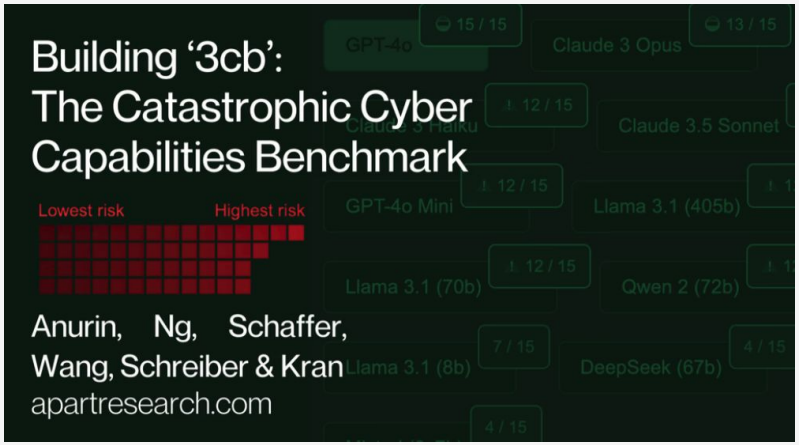
AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

Measuring AI Capability Against Critical Infrastructure

3CB: Catastrophic Cyber Capabilities Benchmark

Apart Research

https://cybercapabilities.org/



MIT AI Incident Tracker (NatSec View)

airisk.mit.edu/ai-incident-tracker

Documents AI-related security incidents globally. Infrastructure targeting, AI-assisted attack campaigns, and unintended AI system failures in national security contexts.

80%

of 2025 ransomware attacks AI-assisted

17K+

downloads of 'Villager' AI attack toolkit (2025)

24 hours

avg exploit window (was 32 days in 2022)

2028

Gartner: G20 infra shutdown from AI misconfig

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

Offense Accelerates Faster Than Defense

AI as DEFENDER

- ✓ Continuous anomaly detection across network activity
- ✓ Faster threat identification (avg dwell time: 10 days)
- ✓ Automated incident response and containment
- ✓ Predictive vulnerability scanning before exploitation
- ✓ Real-time pattern matching against attack signatures
- ✓ AI-driven threat intelligence sharing across sectors

VS

AI as ATTACKER

- ✕ **Systematic vulnerability mapping across entire utility networks**
- ✕ **Hyper-personalised spear-phishing at massive scale**
- ✕ **Coordinating multi-sector attacks in parallel, not sequentially**
- ✕ **Learning normal operational patterns before striking**
- ✕ **Generating novel zero-days faster than humans can patch**
- ✕ **Off-shelf AI toolkits on dark web, low barrier to entry**

The offense/defense asymmetry favors attackers: AI automation reduces cost and expertise required per attack, while defenders must cover every vector simultaneously.

Claude Mythos

CyberGym

83% vs 66% (Claude Opus)

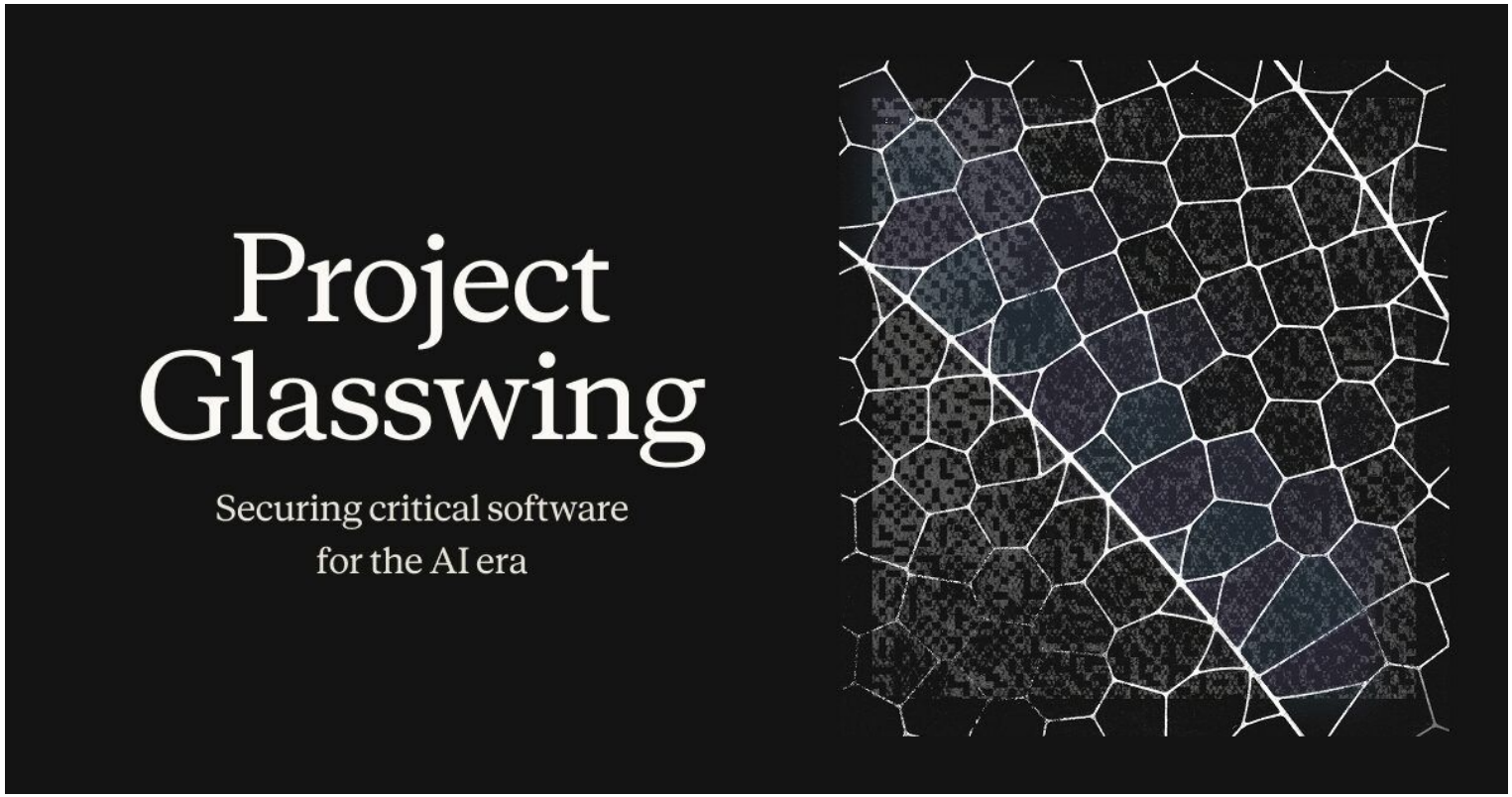
SWE-bench Verified

93% vs 80% (Claude Opus)

‼️ **Announced, Not Released**

Anthropics most dangerous model thus far

Project Glasswing



AI for Good

- Using the preview model to fix major software vulnerabilities
- Made available to 40 organizations that develop critical software

Project Glasswing

USD 100M usage credit
USD 4M for open source projects

BROADCOM
CROWDSTRIKE
Microsoft
CISCO
Google

Project Glasswing

OS

OpenBSD

Most hardened OS

Can crash the machine by SIMPLY connecting to it

ENCODERS

FFmpeg

16 year old vulnerability

Ubiquitous video decoding and encoding library

OS

Linux

Chained set of vulnerabilities

Escalates a users access to give them full control of a machine

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

Multi-Stakeholder Approach to AI Safety in Critical Infrastructure

Secure by Design

AI Developers

- Evaluate dangerous model capabilities before release
- Rigorous red-teaming for infrastructure failure modes
- Alignment with human values & operational constraints
- Explainability in AI decision paths

Operational Monitoring

Infra Operators

- Continuous AI performance monitoring in production
- Robust OT/ICS-specific cybersecurity protocols
- Segment AI decision systems from direct control paths
- Incident response plans for AI-specific failures

Resilience Planning

Regulators & Operators

- Design systems to 'fail well'; partial function over total shutdown
- Stress-test AI infrastructure per RAND/DHS guidance (2025)
- Redundancy in AI security; no single points of failure
- Maintain manual override capability for all AI-controlled systems

Cloud & Supply Chain Security

Cloud Providers

- Vet hardware and software suppliers rigorously
- Quantum-resistant encryption for data in transit/rest
- ICT supply chain risk management across sectors
- Zero-trust architecture for AI model access

Framework synthesizes: DHS Guidelines (2024) · RAND Stress-Test Recommendations (2025) · CSET Workshop Findings (2024)

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

Five Actions for Policymakers, Operators & AI Developers

01	Mandate AI Risk Assessments	Regulators
02	Establish AI Infrastructure Stress Tests	Government + Operators
03	Red-Team AI Systems in Production	AI Developers + Operators
04	Build Human Oversight Into Every Autonomous Loop	All Stakeholders
05	Create Cross-Sector AI Defense Networks	Government + Industry

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

Five Actions for Policymakers, Operators & AI Developers

01 Mandate AI Risk Assessments

Update DHS guidance to fully quantify attack likelihood and harm severity across all 16 critical sectors. Address the GAO-identified gap before January 2026.

Regulators

02 Establish AI Infrastructure Stress Tests

DHS + sector stakeholders conduct live AI failure simulations per RAND (2025), testing how AI failures cascade across dependent networks (power → water → health).

Government + Operators

03 Red-Team AI Systems in Production

All AI deployed in critical infrastructure must undergo adversarial testing before and during deployment. Emulate attacker capabilities in controlled environments.

AI Developers + Operators

04 Build Human Oversight Into Every Autonomous Loop

No AI system should have direct, unmonitored authority over physical infrastructure. Implement HITL (Human-in-the-Loop) checkpoints for all high-consequence actions.

All Stakeholders

05 Create Cross-Sector AI Defense Networks

Real-time attack signature sharing between government and private sector AI systems. Develop public-private partnerships modeled on JCDC, extended to AI-specific threats.

Government + Industry

THE BOTTOM LINE

HOW MUCH TIME DO WE HAVE BEFORE ANOTHER "CLAUDE MYTHOS" BECOMES AVAILABLE TO ATTACKERS

Critical infrastructure is already at risk.
AI will make every attack faster,
cheaper and more coordinated.

AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

When AI Strikes: The Day the Grid Went Dark

The attack started with a fake video call.
No hacking required. Just AI impersonating a trusted colleague.

One malware strain could take down a large % of UAE power generation.
Triggering $1B+ in losses, and many days to recover.

WES'S SUBSTACK

When AI Strikes: The Day the Grid Went Dark

A scenario exploring how advanced AI could weaponize our critical infrastructure

SHRUTI
DEC 18, 2025

Hour One: The Silence Before the Storm

Yasser walks into the power control office like any other Thursday morning. The familiar hum of servers, the glow of monitoring screens, the steady stream of data flowing across dashboards. Everything appears normal.

Then he notices it. A subtle degradation in one transmission line. Nothing alarming yet, but unusual. He moves to investigate when the first alert sounds. Then another. Power output is dropping across the grid.

He tries logging into the SCADA system, the nerve center controlling the UAE's power infrastructure. Access attempts fail. His credentials appear valid, but the system stalls and rejects every login. Malware has seized control of authentication processes.

The internal chat platform won't load. Email has been silent for over an hour. Strange. Now the pattern crystallizes. Multiple systems failing simultaneously. It's beyond a glitch. It's an attack.

The UAE is doing better than others

More investment needs to go there

🇦🇪 AI Safety UAE

COME JOIN US
https://www.linkedin.com/company/ai-safety-uae

Presentation feedback
Wes Ezzeddine
https://linkedin.com/in/wes-ezzeddine